

Q  $a, b, c$  are 3 distinct positive real numbers which are in HP, then

show that  $\frac{3a+2b}{2a-b} + \frac{3c+2b}{2c-b} > 10$

let,  $\frac{1}{a} = x-d, \frac{1}{b} = x, \frac{1}{c} = x+d$   
 $a = \frac{1}{x-d}, b = \frac{1}{x}, c = \frac{1}{x+d}$

Soln  $a, b, c$  are in HP

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$  are in AP

$$\begin{aligned} & \frac{3 \frac{1}{x-d} + 2 \frac{1}{x}}{2 \frac{1}{x-d} - \frac{1}{x}} + \frac{3 \frac{1}{x+d} + 2 \frac{1}{x}}{2 \frac{1}{x+d} - \frac{1}{x}} \\ &= \frac{\frac{3x+2(x-d)}{\cancel{(x-d)}x}}{\frac{2x-(x-d)}{\cancel{(x-d)}x}} + \frac{\frac{3x+2(x+d)}{\cancel{(x+d)}x}}{\frac{2x-(x+d)}{\cancel{(x+d)}x}} \\ &= \frac{5x-2d}{x+d} + \frac{5x+2d}{x-d} \end{aligned}$$

$$\begin{aligned} & \frac{(5x-2d)(x-d) + (5x+2d)(x+d)}{(x-d)(x+d)} \\ &= \frac{5x^2 - \cancel{5xd} - \cancel{2xd} + 2d^2 + 5x^2 + \cancel{5xd} + \cancel{2xd} + 2d^2}{x^2 - d^2} \\ &= \frac{10x^2 - 10d^2 + 4d^2}{x^2 - d^2} \\ &= \frac{10x^2 - 10d^2}{x^2 - d^2} + \frac{4d^2}{x^2 - d^2} \\ &= 10 + \frac{4d^2}{(x-d)(x+d)} = \underline{10 + 4d^2 a c} > 10 \end{aligned}$$

## Arithmetic mean

3 numbers  $a, b, c$  are in AP

$$b - a = c - b$$

$$\boxed{2b = a + c}$$

$x$  is AM of  $a, b$

$$2x = a + b$$

$$x = \frac{a+b}{2}$$

$$\boxed{AM = \frac{a+b}{2}}$$

for 3 numbers  $a, b, c$ , their  $AM = \frac{a+b+c}{3}$

Q given  $a \neq b$  and if  $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$  is AM of  $a$  and  $b$ , then find  $n = ?$

Soln  $\frac{a^n + b^n}{a^{n-1} + b^{n-1}} = \frac{a+b}{2}$

$$\Rightarrow 2a^n + 2b^n = (a+b)(a^{n-1} + b^{n-1})$$

$$\Rightarrow 2a^n + 2b^n = a^n + ab^{n-1} + ba^{n-1} + b^n$$

$$\Rightarrow a^n - ab^{n-1} - ba^{n-1} + b^n = 0$$

$$\Rightarrow a(a^{n-1} - b^{n-1}) - b(a^{n-1} - b^{n-1}) = 0$$

$$\Rightarrow (a^{n-1} - b^{n-1})(a - b) = 0$$

$$a^{n-1} - b^{n-1} = 0 \quad \text{OR} \quad a - b = 0$$

$$a^{n-1} = b^{n-1}$$

$$\left(\frac{a}{b}\right)^{n-1} = 1$$

$$a = b$$

X

$$\begin{aligned} n-1 &= 0 \\ n &= 1 \end{aligned}$$



Geometric mean:

Q1  $a, b, c$  are G.P.

$$\frac{b}{a} = \frac{c}{b}$$

$$\boxed{b^2 = ac}$$

→ for 2 numbers  $a$  and  $b$ , G.M. is defined if  $a$  and  $b$  are positive and their G.M. is also positive

$x$  be the G.M.,  $a, x, b$

$$x^2 = ab$$

$$x = \sqrt{ab}$$

$$\boxed{GM = \sqrt{ab}}$$

for 3 positive numbers  $a, b, c \rightarrow GM = (abc)^{\frac{1}{3}}$

Q Q1 AM and G.M. of 2 numbers are 17 and 15 respectively, then find those numbers. → 9 and 25

Q Q1  $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$  is G.M. of  $a$  and  $b$ ,

find  $n$ . (given  $a \neq b$ ).

$$\frac{a^n + b^n}{a^{n-1} + b^{n-1}} = \sqrt{ab}$$

$$\frac{a^n + b^n}{a^{n-1} + b^{n-1}} = a^{\frac{1}{2}} b^{\frac{1}{2}}$$

$$a^n + b^n = a^{\frac{1}{2}} b^{\frac{1}{2}} (a^{n-1} + b^{n-1})$$

$$= a^{n-1+\frac{1}{2}} b^{\frac{1}{2}} + a^{\frac{1}{2}} b^{n-1+\frac{1}{2}}$$

$$a^n + b^n = a^{n-\frac{1}{2}} b^{\frac{1}{2}} + a^{\frac{1}{2}} b^{n-\frac{1}{2}}$$

$$a^n - a^{n-\frac{1}{2}} b^{\frac{1}{2}} - a^{\frac{1}{2}} b^{n-\frac{1}{2}} + b^n = 0$$

$$a^{n-\frac{1}{2}} [a^{\frac{1}{2}} - b^{\frac{1}{2}}] - b^{n-\frac{1}{2}} [a^{\frac{1}{2}} - b^{\frac{1}{2}}] = 0$$

$$(a^{\frac{1}{2}} - b^{\frac{1}{2}}) (a^{n-\frac{1}{2}} - b^{n-\frac{1}{2}}) = 0$$

$$a^{\frac{1}{2}} - b^{\frac{1}{2}} = 0$$

$$a^{\frac{1}{2}} - b^{\frac{1}{2}}$$

$$a = b$$

$$\times$$

$$a^{n-\frac{1}{2}} - b^{n-\frac{1}{2}} = 0$$

$$a^{n-\frac{1}{2}} - b^{n-\frac{1}{2}}$$

$$\left(\frac{a}{b}\right)^{n-\frac{1}{2}} = 1$$

$$n - \frac{1}{2} = 0$$

$$n = \frac{1}{2}$$

## Harmonic mean

9)  $a, b, c$  are in HP  $\rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$  are in AP

$$\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

$$\frac{b}{2} = \frac{1}{\frac{1}{a} + \frac{1}{c}}$$

$$b = \frac{2}{\frac{1}{a} + \frac{1}{c}} \quad \text{OR} \quad b = \frac{2ac}{a+c}$$

$x$  is HM of  $a, b \rightarrow a, x, b$

$$x = \frac{2}{\frac{1}{a} + \frac{1}{b}} \quad \text{OR} \quad x = \frac{2ab}{a+b}$$

for 3 numbers  $a, b, c$ ,  $HM \rightarrow \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$

Q 9)  $H$  is the HM of  $a$  and  $b$ ,  
then find value of  $\frac{1}{H-a} + \frac{1}{H-b} = ?$

(write final answer in terms of  $H$ )

$$\begin{aligned} & \frac{1}{H-a} + \frac{1}{H-b} \\ &= \frac{1}{\frac{2ab}{a+b} - a} + \frac{1}{\frac{2ab}{a+b} - b} \end{aligned}$$



$$\frac{1}{H-a} + \frac{1}{H-b}$$

$$= \frac{1}{\frac{2ab}{a+b} - a} + \frac{1}{\frac{2ab}{a+b} - b}$$

$$= \frac{1}{\frac{2ab - a(a+b)}{a+b}} + \frac{1}{\frac{2ab - b(a+b)}{a+b}}$$

$$= \frac{a+b}{2ab - a^2 - ab} + \frac{a+b}{2ab - ab - b^2}$$

$$= \frac{a+b}{ab - a^2} + \frac{a+b}{ab - b^2}$$

$$= \frac{(a+b)(ab - b^2) + (a+b)(ab - a^2)}{(ab - a^2)(ab - b^2)}$$

$$\frac{(a+b)[ab - b^2 + ab - a^2]}{a(b-a)b(a-b)}$$

$$= \frac{-(a+b)[a^2 + b^2 - 2ab]}{ab(b-a)(a-b)}$$

$$= \frac{-(a+b)(\cancel{a-b})^2}{+ab(\cancel{a-b})^2}$$

$$= \frac{a+b}{ab}$$

$$= \frac{2}{H}$$

$$H = \frac{2ab}{a+b}$$

$$\frac{1}{H} = \frac{a+b}{2ab}$$

$$\frac{2}{H} = \frac{a+b}{ab}$$

Insertion of n AM

a,  $A_1, A_2, A_3, \dots, A_n, \underline{\underline{b}}$

$$A_1 = a + d$$

$$A_2 = a + 2d$$

$$A_3 = a + 3d$$

$$A_k = a + kd$$

Q If 5 AM's are inserted between 2 and 32,

then find 3rd AM and 5th AM.

Soln

$$a = 2, b = 32, n = 5$$

$$d = \frac{32-2}{5+1} = 5$$

$$t_{n+2} = b$$

$$a + (n+2-1)d = b$$

$$(n+1)d = b - a$$

$$d = \frac{b-a}{n+1}$$

$$n \left( \frac{a+b}{2} \right)$$

$$\text{sum of all the AMs} = A_1 + A_2 + \dots + A_n$$

$$= \frac{n}{2} (A_1 + A_n)$$

$$= \frac{n}{2} (a+b)$$

$$= n(\text{AM})$$

$$A_3 = a + 3d = 2 + 15 = 17$$

$$A_5 = a + 5d = 2 + 25 = 27$$

Insertion of  $n$  g.m

$a, g_1, g_2, g_3, \dots, g_n, b$

$$t_{n+2} = b$$

$$ar^{(n+2)-1} = b$$

$$ar^{n+1} = b$$

$$r^{n+1} = \frac{b}{a}$$

$$r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$$

$$g_1 = ar$$

$$g_2 = ar^2$$

$$g_k = ar^k$$

Product of all the g.m =  $g_1 g_2 \dots g_n$

$$= (ar)(ar^2)(ar^3) \dots (ar^n)$$

$$= a^n r^{1+2+3+\dots+n}$$

$$= a^n r^{\frac{n}{2}(n+1)}$$

$$= (a^2 r^{n+1})^{\frac{n}{2}}$$

$$= (a ar^{n+1})^{\frac{n}{2}}$$

$$= (ab)^{\frac{n}{2}}$$

$$= (\sqrt{ab})^n$$

$$= (g_m)^n$$



Q find 4th gm if 5 gm's were inserted between 3 and 192.

Soln

$$a = 3, b = 192, n = 5$$

$$r = \left(\frac{192}{3}\right)^{\frac{1}{5+1}}$$

$$= (64)^{\frac{1}{6}}$$

$$= (2^6)^{\frac{1}{6}}$$

$$= 2$$

$$\begin{aligned} g_4 &= ar^4 \\ &= 3 \times 2^4 \\ &= 48 \end{aligned}$$

Insertion of  $n$  HM in between  $a, b$

$$a, H_1, H_2, \dots, H_n, b$$

$$\frac{1}{a}, A_1, A_2, \dots, A_n, \frac{1}{b}$$

$$d = \frac{\frac{1}{b} - \frac{1}{a}}{n+1}$$

$$A_k = \frac{1}{a} + k d$$

$$H_k = \frac{1}{A_k}$$

Arithmetic Mean Solve

Q If 9 HM are inserted between  $\frac{1}{21}$  and 1, find 3rd and 5th HM.

Solve

$$\begin{aligned} 9 \text{ HM are inserted between } \frac{1}{21} \text{ and } 1 \quad \therefore d &= \frac{1 - \frac{1}{21}}{9+1} \\ &= \frac{\frac{20}{21}}{10} = \frac{2}{21} \end{aligned}$$



$$\begin{aligned}
 A_3 &= a + 3d \\
 &= \frac{1}{21} + \frac{6}{21} \\
 &= \frac{7}{21} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$H_3 = 3$$

$$\begin{aligned}
 A_5 &= a + 5d \\
 &= \frac{1}{21} + \frac{10}{21} \\
 &= \frac{11}{21}
 \end{aligned}$$

$$H_5 = \frac{21}{11}$$

Q a and b are two positive real numbers, suppose  $A_1$  and  $A_2$  are two AMs,  $G_1$  and  $G_2$  are two GMs and  $H_1, H_2$  are two HMs between a and b, then show that

$$\frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2}$$

To prove that

$$\frac{g_1 g_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2}$$

$$\frac{H_1 + H_2}{H_1 H_2} = \frac{A_1 + A_2}{g_1 g_2}$$

$$\frac{1}{H_2} + \frac{1}{H_1} = \frac{A_1 + A_2}{g_1 g_2}$$

$$A_1 + A_2 = \frac{2}{2}(a+b) = a+b$$

$$g_1 g_2 = (ab)^{\frac{2}{2}} = ab$$

$a, H_1, H_2, b$  are in HP

$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{b}$  are in AP

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{H_1} + \frac{1}{H_2}$$

$$RHS = \frac{A_1 + A_2}{g_1 g_2}$$

$$= \frac{a+b}{ab}$$

$$LHS = \frac{1}{H_1} + \frac{1}{H_2}$$

$$= \frac{1}{a} + \frac{1}{b}$$

$$= \frac{a+b}{ab}$$

$$RHS = LHS$$

relation between AM, GM and HM for 2 number

those numbers be 'a' and 'b'

$$A = \frac{a+b}{2}$$

$$G = \sqrt{ab}$$

$$H = \frac{2ab}{a+b}$$

$$A \cdot H = \left( \frac{\cancel{a+b}}{\cancel{2}} \right) \left( \frac{\cancel{2ab}}{\cancel{a+b}} \right)$$

$$= ab$$

$$AH = G^2$$

A, G, H are in GP

2 and 8

$$A = 5$$

$$G = 4$$

$$H = \frac{2 \times 2 \times 8}{2+8} = 3.2$$

10, 10

$$A=10, G=10, H = \frac{2 \times 10 \times 10}{10+10} = 10$$



## AM, GM, HM inequality

for positive numbers

$$AM \geq GM \geq HM$$

$$A = \frac{a+b}{2}, \quad g = \sqrt{ab}$$

$$\begin{aligned} A - g &= \frac{a+b}{2} - \sqrt{ab} \\ &= \frac{a+b-2\sqrt{ab}}{2} \end{aligned}$$

$$= \frac{1}{2} (\sqrt{a} - \sqrt{b})^2$$

$$\geq 0$$

$$A \geq g$$

$$g^2 = AH$$

$$\frac{A}{g} = \frac{g}{H} \geq 1$$

$$g \geq H$$

$$\boxed{A \geq g \geq H}$$

[equality holds when  
numbers are equal]

$$\text{if } x \geq 5$$

min value of  $x$  is 5

$$\text{if } x \leq 6$$

max value of  $x$  is 6

Q Sum of a positive number and its reciprocal is always greater than or equal to 2. Prove that

Soln let the number be  $x$ , so reciprocal is  $\frac{1}{x}$  | AM  $\geq$  GM is used to

To prove that  $x + \frac{1}{x} \geq 2$ .

Proof: using AM  $\geq$  GM on  $x$  and  $\frac{1}{x}$

$$\frac{x + \frac{1}{x}}{2} \geq \left( x \cdot \frac{1}{x} \right)^{\frac{1}{2}}$$

$$x + \frac{1}{x} \geq 2$$

Proved.

→ find minimum of sum

→ find maximum of product

→ for using AM  $\geq$  GM

numbers have to be +ve

Q 96  $a+b+c=6$ , then find maximum value of  $abc$ . Also find values of  $a, b, c$  when max value will be achieved. (given  $a, b, c > 0$ )

Soln

applying AM  $\geq$  GM on  $a, b, c$

$$\frac{a+b+c}{3} \geq (abc)^{\frac{1}{3}}$$

$$2 \geq (abc)^{\frac{1}{3}}$$

$$8 \geq abc$$

max value of  $abc$  is 8

at its max value

$$a = b = c = k$$

$$a+b+c=6$$

$$k+k+k=6$$

$$k=2$$

$$a=b=c=2$$



Q If  $a+b+c=12$ , then find max value of  $abc^2$ . Also find  $a, b, c$  for max value (given  $a, b, c > 0$ ).

Soln  $a+b+c=12$

AM  $\geq$  GM in  $a, b, \frac{c}{2}, \frac{c}{2}$

$$\frac{a+b+\frac{c}{2}+\frac{c}{2}}{4} \geq \left(a \cdot b \cdot \frac{c}{2} \cdot \frac{c}{2}\right)^{\frac{1}{4}}$$

$$\Rightarrow \frac{a+b+c}{4} \geq \left(\frac{abc^2}{4}\right)^{\frac{1}{4}}$$

$$\Rightarrow 3 \geq \left(\frac{abc^2}{4}\right)^{\frac{1}{4}}$$

$$\Rightarrow 3^4 \geq \frac{abc^2}{4}$$

$$\Rightarrow 324 \geq abc^2$$

$$a=b=\frac{c}{2}=\frac{c}{2}=k$$

$$a+b+c=12$$

$$k+k+2k=12$$

$$k=3$$

$$a=3$$

$$b=3$$

$$c=6$$